

AI-Powered Unified Cyber-Fraud Intelligence Platform

Explainable AI & Quantum Risk Monitoring | Finspark Hackathon 2026

Team Name : [Add Your Team Name]

Team Bio : [1-2 lines about your team]

Date : [Add Submission Date]

Problem Statement?

Cybersecurity intrusions and financial fraud are converging into a single attack chain — but banks still defend them in separate silos.

- **Multi-stage attacks:** credential theft and session hijacking (a cyber signal) precede a fraudulent transfer (a fraud signal) within minutes; unimodal SIEM and fraud engines see neither half.
- **\$11.88 trillion** — the projected global cost of cybercrime in 2026, heading toward \$19.71 trillion by 2030.
- **85% YoY rise** in India's UPI fraud complaints — over 2.4 million cases and ₹22,495 crore in losses reported in 2025 alone.
- **120 seconds:** the time stolen funds take to disperse through 4–7 mule accounts after an Authorized Push Payment (APP) scam.
- **Harvest-Now-Decrypt-Later:** adversaries are archiving encrypted banking traffic today to decrypt retroactively once quantum computers mature.

We chose this problem because point solutions — SIEM, EDR, fraud engines — each see only half the attack. Closing that gap needs one fused, explainable, quantum-ready platform.

Pre-Requisite

Building and evaluating the platform requires the following inputs:

- **Cyber telemetry:** SIEM/EDR logs, Active Directory & OAuth authentication records, API access tokens, firewall/VPN logs.
- **Transaction data:** Core Banking System (CBS) ledgers, SWIFT/ACH/UPI payment corridors, device and behavioral biometrics.
- **Benchmark datasets:** PaySim (6.3M synthetic transactions), Elliptic Bitcoin graph dataset, UNSW-NB15 / CICIDS2017 intrusion sets.
- **Infrastructure access:** Kafka/Flink streaming cluster, Neo4j graph database, Kubernetes environment, GPU capacity for GNN training.
- **Regulatory inputs:** RBI DPSC & Master Directions on Fraud, DPDP Act 2023/Rules 2025, CERT-In 6-hour reporting mandate, NIST PQC standards.
- **Data handling:** NDAs and licensing for proprietary bank data; PII hashed or tokenized before ingestion into the research environment.

Tools or resources

A hybrid, production-grade technology stack powers the platform:

- **Frontend:** React.js, Tailwind CSS, D3.js for interactive, physics-based threat graphs.
- **Backend:** Go for high-concurrency ingestion microservices; Python (FastAPI) for the ML and graph-analytics layer.
- **Streaming:** Apache Kafka for event ingestion, Apache Flink for in-flight ISO 20022 / SWIFT MT validation.
- **Databases:** PostgreSQL (relational), Neo4j (graph relationships), Redis (low-latency feature cache).
- **ML & XAI:** XGBoost, LightGBM, Isolation Forest, GraphSAGE (PyTorch/DGL), Autoencoders, SHAP, LIME, GNNExplainer.
- **MLOps & Infra:** MLflow, Seldon Core, Docker, Kubernetes, Terraform, Prometheus/Grafana.
- **Datasets & references:** PaySim, Elliptic Bitcoin, UNSW-NB15/CICIDS2017, NIST PQC (ML-KEM/ML-DSA), RBI & DPDP guidelines.

Supporting Functional Documents

The design is grounded in the following research and compliance references:

- **Research dossier:** “AI-Powered Unified Cybersecurity & Financial Fraud Intelligence Platform” — architecture, ML pipeline and database schema.
- **Whitepaper:** “Securing the Future of Banking” — AI-driven correlation of cyber telemetry and transactional behaviour.
- **Regulatory mapping:** RBI Digital Payment Security Controls, DPDP Act 2023 & Rules 2025, CERT-In IT Rules 2022 (6-hour reporting).
- **Cryptographic standards:** NIST FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), FIPS 205 (SLH-DSA).
- **Draft IEEE paper:** “Domain Fusion — Correlating Cybersecurity Telemetry and Financial Fraud Signatures via Heterogeneous GNNs.”
- **System design:** complete architecture, ER diagram and API contracts — see Slide 14.

Key Differentiators & Adoption Plan

What sets this platform apart — and how institutions can adopt it:

- **Domain fusion:** one Heterogeneous GNN correlates cyber events with transaction risk in real time, not after the fact.
- **Quantum-risk native:** SOC Quantum Playbook with 5 detection use cases for HNDL and hybrid-downgrade attacks.
- **Layered XAI:** SHAP (global), LIME (local), GNNExplainer (subgraphs) — routed automatically by transaction risk tier.
- **Compliance-by-design:** automated RBI DPSC, DPDP consent workflows and CERT-In 6-hour incident reporting.

Adoption Plan

- **Phase 1:** pilot on one branch or product line using existing SIEM + CBS feeds.
- **Phase 2:** expand ingestion to enterprise-wide Kafka streaming and graph modeling.
- **Phase 3:** activate XAI, quantum monitoring and compliance automation.
- **Phase 4:** extend to federated learning across branches and partner institutions.

GitHub Repository Link & supporting diagrams, screenshots, if any

Prototype repository and supporting references:

- **GitHub Repository:** [Add your repository link here]
- **Architecture Diagram:** the full end-to-end system design is on Slide 14.
- **Screenshots & Demo Video:** prototype visuals and walkthrough video are on Slide 15.
- **Research Dossier & Whitepaper:** the underlying architecture and compliance research referenced throughout this deck.

Business Potential and Relevance

A large and urgently growing market:

- **Market size:** global cybercrime cost of \$11.88 trillion in 2026, projected to reach \$19.71 trillion by 2030.
- **Rising spend:** global cybersecurity spending climbing from \$227B (2025) to \$308B (2026).
- **Target customers:** Tier-1/2 commercial banks, payment gateways, Non-Banking Financial Companies, crypto exchanges.
- **Regulatory tailwind:** the DPDP Act Consent Manager deadline (Nov 2026) and RBI DPSC mandates force legacy-system replacement.
- **Long-term applicability:** extensible to insurance fraud, cross-border correlation and cross-bank federated intelligence sharing.

Uniqueness of Approach and Solution

Existing tools solve half the problem — we solve the whole one:

- **The gap:** SIEMs (Splunk, Sentinel) read IT telemetry but not transactions; fraud platforms (Feedzai, Mastercard DI) see transactions but not the cyber precursor.
- **Our approach:** a Heterogeneous GraphSAGE model unifies accounts, devices, IPs and transactions into one evolving graph.
- **Adaptive explainability:** an Explanation Strategy Router tailors XAI depth to risk — cached scores for low-risk, full SHAP/GNNExplainer for high-risk alerts.
- **Privacy-preserving intelligence:** Federated Learning with differential privacy ($\epsilon=1.5$, $\delta=10^{-5}$) detects cross-institution fraud rings without sharing raw data.
- **Quantum-first:** a native SOC Quantum Playbook monitors post-quantum downgrade attacks — a capability no competitor product offers today.

User Experience

Designed to cut alert fatigue across 12-hour SOC shifts:

- **Single pane of glass:** SIEM timeline (left), transaction ledger (right), fused risk score and XAI explanation (center).
- **Dark UI:** red/amber/green severity coding reduces analyst eye strain over long shifts.
- **Threat Graph Visualizer:** click a suspected mule account to instantly expand linked devices, IPs and transfer destinations.
- **Investigation Copilot:** an LLM+RAG assistant answers analyst questions and cites the exact policy or case file behind a decision.
- **Counterfactual explanations:** tell analysts precisely what would have kept a transaction within the ALLOW threshold.

Scalability

Engineered for tens of thousands of transactions per second:

- **Elastic streaming:** Apache Kafka + Kubernetes horizontal scaling handle enterprise-wide log and transaction volume.
- **Inductive graphs:** GraphSAGE generates embeddings for new accounts and devices instantly — no full graph retraining.
- **Tiered scoring:** LightGBM scores inline in under 10ms; deep graph traversals run asynchronously off pre-computed embeddings.
- **Load-tested:** validated to sustain 10,000+ TPS under simulated peak (“Black Friday”) transaction volumes.
- **Cross-institution scale:** Federated Learning lets the same model scale across branches and partner banks without centralizing data.

Ease of Deployment and Maintenance

Cloud-native and self-service from day one:

- **Containerized:** microservices run on Docker, orchestrated by Kubernetes with Horizontal Pod Autoscaling.
- **Automated pipelines:** Infrastructure as Code via Terraform; CI/CD via GitHub Actions with embedded SAST/DAST scanning.
- **MLOps:** MLflow and Seldon Core manage model versioning, A/B testing and rolling redeployment.
- **Observability:** Prometheus/Grafana and the ELK stack provide continuous monitoring and alerting.
- **Rollout plan:** a 12-month, 5-phase roadmap — Foundation → Ingestion → AI/Graph → Explainability → Quantum & Compliance.

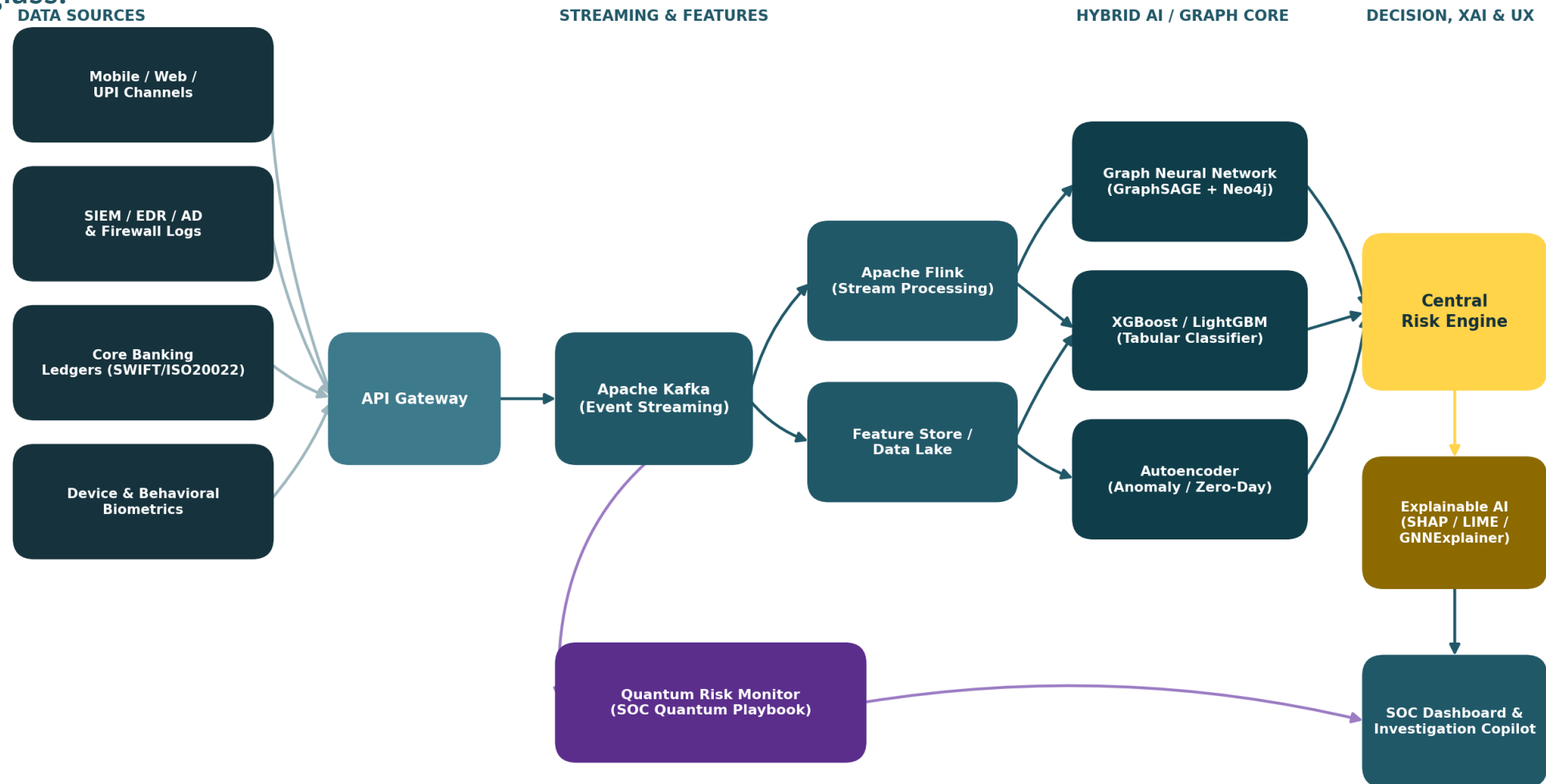
Security Considerations

Security is embedded at every layer, not bolted on:

- **Zero Trust:** mutual TLS between services, OIDC plus mandatory MFA for all administrative access.
- **Encryption:** AES-256 at rest, TLS 1.3 in transit, migrating to hybrid ML-KEM key exchange.
- **Post-quantum readiness:** SOC Quantum Playbook covers hybrid-downgrade detection, certificate lifecycle anomalies and HNDL indicators.
- **Compliance-native:** DPDP Act consent manager APIs, RBI DPSC device binding & data localization, CERT-In 6-hour incident reporting.
- **Model integrity:** checkpoints hashed with SHA-256 and anchored on a permissioned blockchain, signed with post-quantum CRYSTALS-Dilithium.

Architecture diagram & Images

Kafka-driven ingestion → hybrid GNN / XGBoost / Autoencoder core → explainable, quantum-aware decisioning on a single pane of glass.



Solution Screenshots & video & Github Link

Prototype artifacts to accompany the live demo:

- **GitHub Repository:** [Add your repository link here]
- **Demo Video:** [Add your demo video link here]
- **SOC Dashboard screenshot:** Global Threat Level, TPS, Active Alerts, and 24-hour fraud intercepted.
- **Investigation Screen screenshot:** SIEM timeline, transaction ledger and XAI explanation panel side by side.
- **Threat Graph Visualizer screenshot:** an expanded mule-account network showing linked devices and transfer destinations.
- **Quantum Risk Dashboard screenshot:** live cryptographic inventory showing PQC migration progress.

Thank You

- [Team Member 1 — Role — Email/Phone]
- [Team Member 2 — Role — Email/Phone]
- [Team Member 3 — Role — Email/Phone]
- [Team Member 4 — Role — Email/Phone]

Thank you for the opportunity to present at Finspark Hackathon 2026. We welcome your questions.